

BIOL 40C

Midterm 2 Practice

+ Cumulative Concept Map

Monday, May 11, 2026 — Class 11

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How Today Works

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AGENDA

10:00-10:05	Frame the day
10:05-10:55	TIMED PRACTICE (50 min, exam conditions)
10:55-11:05	Break
11:05-11:35	Team review of every answer
11:35-11:45	Concept map activity (10 min build)
11:45-11:50	Q&A + Wednesday logistics

The Practice — Ground Rules

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PRACTICE

Treat the next 50 minutes as if they were the real thing.

- Phones in your bag. Watches off. No notes, no laptops.
- 20 multiple choice + 3 short answer + 1 clinical case.
- If you finish early: REREAD your work. Do not leave.
- The practice is NOT graded. Its job is to find your weak spots.
- Be honest with yourself: the questions you guessed are tonight's homework.

Wednesday's exam: ~25 MCQ + 4 short answer + 1 case, full 1 h 50 min. Same difficulty as today.

PRACTICE EXAM IN PROGRESS

50:00

(Use a real timer — this slide is just a placeholder.)

Heads down. No talking. You've got this.

Answer Key — MCQ 1-20

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ANSWER KEY

1. c	2. b	3. b	4. b
5. c	6. a	7. b	8. b
9. a	10. b	11. b	12. a
13. b	14. a	15. d	16. b
17. a	18. d	19. c	20. c

Show of hands. We'll walk through any question more than 3 hands missed.

Reteach 1: ACE Inhibitors and GFR (Q11)

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RETEACH

ACE inhibitor → blocks angiotensin II formation

Mechanism

Renin → AT-I → (ACE) → AT-II

AT-II constricts the EFFERENT arteriole

ACE-i blocks step 3

→ Efferent dilates

→ Glomerular hydrostatic pressure FALLS

→ NFP falls

→ GFR falls

Clinical takeaway

Why we still use them in Reyna (diabetic nephropathy):

Lowering glomerular pressure PROTECTS the damaged filter from further injury. She loses a little GFR up front, but loses much LESS over the long run.

Same drug, same mechanism, two different framings.

Reteach 2: Body Fluid Compartments (Q15)

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RETEACH

Total body water (TBW) $\approx$ 60% of body weight. $\sim$ 42 L in a 70 kg adult.

ICF

Intracellular Fluid

$\sim$ **25 L**

$\approx$ 2/3 of TBW

High K^+ , high phosphate, low Na^+

ECF — Interstitial

$\sim$ 12 L ($\approx$ 80% of ECF)

Juice between cells

ECF — Plasma

$\sim$ 3 L ($\approx$ 20% of ECF)

ONLY a quarter of ECF

"The blood lost 1 L" $\rightarrow$ picture a tiny straw out of a big tank.

Reteach 3: IV Fluid Choice (Q18)

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RETEACH

Normal Saline

0.9% NaCl

308 mOsm/L

Stays in ECF

USE FOR:

ECF volume resuscitation (hemorrhage, GI losses, dehydration)

D5W

5% dextrose in water

Effectively free water once dextrose is metabolized

Distributes 2/3 ICF, 1/3 ECF

USE FOR:

Replacing FREE-water deficits (e.g., hyponatremia)

Lactated Ringer's

Isotonic + electrolytes

~ 273 mOsm/L; has K^+ , Ca^{2+} , lactate $\rightarrow$ bicarb

Stays in ECF

USE FOR:

Trauma, surgery, burns. AVOID in CKD (K^+ load).

The trick: "Volume deficit?" $\rightarrow$ NS or LR. "Pure water deficit?" $\rightarrow$ D5W.

Reteach 4: The 2x2 ABG Matrix (Q19)

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RETEACH

	Low pH (Acidosis)	High pH (Alkalosis)
PCO₂ driver	Respiratory Acidosis ↑ PCO ₂ (CO ₂ retention) COPD, opioid overdose	Respiratory Alkalosis ↓ PCO ₂ (hyperventilation) panic, salicylate, sepsis
HCO₃⁻ driver	Metabolic Acidosis ↓ HCO ₃ ⁻ DKA, lactic, diarrhea, uremia (check anion gap)	Metabolic Alkalosis ↑ HCO ₃ ⁻ vomiting, aldosteronism (Oscar)

30-sec algorithm: pH? CO₂? HCO₃⁻? If pH and PCO₂ move opposite → respiratory; if pH and HCO₃⁻ move same → metabolic.

Reteach 5: Aldosterone & The Principal Cell (Q20)

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RETEACH

LUMEN

BLOOD

Principal Cell

ENaC →

Na⁺ in (apical)

ROMK ↗

K⁺ out into lumen (down lumen-negative voltage)

Na⁺/K⁺ ATPase

Na⁺ out → blood, K⁺ in (basolateral)

H⁺ ATPase

H⁺ pumped into lumen (intercalated cells nearby)

Aldosterone UPREGULATES all four →

If aldosterone ↑: Na⁺ ~ unchanged (water follows), K⁺ ↓, H⁺ ↓ in blood = pH ↑, BP ↑. All four of Oscar's findings.

Short Answer + Case (Carlos)

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ANSWER KEY

Talk through these in your team. The case is what Wednesday will reward most.

Short Answer Highlights

SA-1 (diarrhea): NAGMA. Loss of HCO_3^- in stool, Cl^- rises, AG normal.

SA-2 (loop diuretic): $\uparrow$ urine, $\downarrow$ BP, $\downarrow$ K^+ (via $\uparrow$ distal Na^+ delivery $\rightarrow$ aldosterone).

SA-3: ADH = osmolality + free water. Aldosterone = volume + Na^+ .

Carlos = DKA

HAGMA from ketoacids (β -hydroxybutyrate, acetoacetate). Kussmaul = respiratory comp. K^+ falsely high (insulin def + acidosis), will crash with treatment. Start NS, then insulin once $\text{K}^+ \geq 3.3$. Elevated BUN/Cr = dehydration.

That single case integrates digestive $\rightarrow$ metabolism $\rightarrow$ urinary $\rightarrow$ fluid/acid-base. Wednesday's case will look similar in shape.

Concept Map: One Body, Four Systems

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ACTIVITY

10 minutes. Teams of 3-4. One sheet of large paper, colored markers.

DIGESTIVE

Wk 0-1

Patient: Juan (celiac)

List 3-5 concepts. Add an arrow to next box.

METABOLISM

Wk 2

Patient: Brown Bag patients

List 3-5 concepts. Add an arrow to next box.

FLUID/AB

Wk 4

Patient: Reyna + Oscar

List 3-5 concepts. Add an arrow to next box.

URINARY

Wk 3

Patient: Santiago (HUS)

List 3-5 concepts. Add an arrow to next box.

Use a 5th color for concepts that show up in MORE than one box (Na^+/K^+ ATPase, osmosis, feedback loops). Those are the threads.

Wednesday — Midterm Logistics

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LOGISTICS

When:	Wed May 13, 10:00–11:50 AM
Where:	Room 8403 (same as today)
Format:	~25 MCQ + 4 short answer + 1 clinical case
Coverage:	Cumulative across Weeks 0–4. Digestive, metabolism, urinary, fluid/acid-base.
Bring:	#2 pencil + backup, pen, Foothill ID, water
Leave home:	Phones, smart watches, calculators, hats with brims, notes
If sick:	Email me before 9 AM Wed. We will arrange a make-up.
DRC accommodations:	Email me by 6 PM today if not already confirmed.

You have done the work.

Sleep tonight. Light study tomorrow.

Wednesday, 10:00 AM, Room 8403.

"Trust the framework you have built. Five patients walked you through it. Now you carry it forward."